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Jeans-Independent Holdback Bypass for AMR Refinement in Cosmological Simulations

Juhan Kim[1]

[1] *Korea Institute for Advanced Study (KIAS), 85 Hoegiro, Dongdaemun-gu, Seoul 02455, Republic of Korea*

ABSTRACT

Cosmological AMR simulations employ a *holdback* mechanism that progressively releases refinement levels as the universe expands, preventing premature over-refinement at high redshift. However, this mechanism blocks *all* refinement criteria simultaneously, including the Jeans length criterion, which is a numerical stability requirement rather than merely a resolution preference. When a holdback level is finally released, a large number of cells suddenly satisfy the refinement condition, triggering an abrupt starburst. We implement a `jeans_bypass_holdback` option that allows the Jeans criterion to operate independently of the holdback, with a configurable level limit (`n_jeans_bypass_levels`) to prevent uncontrolled refinement cascades. We discuss the interplay between the Jeans criterion, the polytropic equation of state, and the density threshold `d_jeans_thre`, showing that the polytropic temperature floor provides a natural self-regulation mechanism for the bypass.

Keywords: methods: numerical — cosmology: simulations — stars: formation — hydrodynamics

1. AMR REFINEMENT IN RAMSES

In RAMSES, each cell at level $\ell < N_{\rm levelmax}$ is tested for refinement at every coarse time-step. The decision to subdivide a cell into $2^{N_{\rm dim}}$ children is governed by a combination of physical, geometric, and cosmological criteria.

1.1. *Refinement Logic*

The overall refinement flag for a cell is determined by the following Boolean expression:

$$\mathcal{R} = \neg\,\mathcal{H} \;\wedge\; \mathcal{G} \;\wedge\; \big(\mathcal{M} \vee \mathcal{S} \vee \mathcal{D} \vee \mathcal{P} \vee \mathcal{U} \vee \mathcal{J}\big), \qquad (1)$$

where each symbol is defined below.

1.1.1. *Physical Criteria (OR)*

1. **Quasi-Lagrangian mass criterion ($\mathcal{M}$).** The cell is flagged if its gas density exceeds a level-dependent threshold:

$$\rho_{\rm cell} \;\geq\; m_{\rm refine}(\ell)\,\frac{m_{\rm sph}}{V_{\rm cell}}, \qquad (2)$$

where $m_{\rm sph}$ is the effective SPH kernel mass and $V_{\rm cell} = (\Delta x_\ell)^3$. This criterion ensures that regions containing roughly $m_{\rm refine}$ or more particles per cell are refined, providing a quasi-Lagrangian resolution that tracks gravitational collapse. Controlled by the namelist array `m_refine(1:nlevelmax)`.

2. **Sink particle forcing ($\mathcal{S}$).** If `sink_refine=.true.`, any cell containing a sink cloud particle (identified by `idp` < 0 and `tp` $= 0$) is unconditionally flagged up to $N_{\rm levelmax}$, guaranteeing maximum resolution around each accretion region.

3. **Density gradient ($\mathcal{D}$).** The normalised density gradient between a cell and each of its $2N_{\rm dim}$ face-neighbours is computed as

$$\epsilon_\rho = 2\,\max\!\left(\frac{|\rho_+ - \rho_0|}{|\rho_+ + \rho_0| + \rho_{\rm floor}},\; \frac{|\rho_0 - \rho_-|}{|\rho_0 + \rho_-| + \rho_{\rm floor}}\right). \qquad (3)$$

If $\epsilon_\rho >$ `err_grad_d` along any axis, the cell is flagged.

4. **Pressure gradient ($\mathcal{P}$).** Analogous to the density gradient but applied to the thermal pressure:

$$\epsilon_P = 2\,\max\!\left(\frac{|P_+ - P_0|}{|P_+ + P_0| + P_{\rm floor}},\; \frac{|P_0 - P_-|}{|P_0 + P_-| + P_{\rm floor}}\right), \qquad (4)$$

flagging when $\epsilon_P >$ `err_grad_p`.

5. **Velocity gradient ($\mathcal{U}$).** A normalised velocity jump relative to the local sound speed:

$$\epsilon_v = 2\,\max\!\left(\frac{|v_+ - v_0|}{c_{s,+} + c_{s,0} + |v_+| + |v_0| + v_{\rm floor}},\; \frac{|v_0 -}{c_{s,0} + c_{s,-} + |v_0}\right. \qquad (5)$$

flagging when $\epsilon_v >$ `err_grad_u` in any component.

6. **Jeans length ($\mathcal{J}$).** When `jeans_refine(`ℓ`)` $= N_J > 0$, the cell is flagged if the local Jeans length is shorter than N_J cell widths:

$$\lambda_{\rm J} = \sqrt{\frac{\pi\, c_s^2}{G\,\rho}} < N_J\,\Delta x_\ell, \tag{6}$$

implementing the Truelove et al. (1997) condition to prevent artificial fragmentation in self-gravitating gas. An additional density threshold `d_jeans_thre` restricts this criterion to cells with $n_{\rm H} \geq$ `d_jeans_thre`.

1.1.2. *Geometry Mask (AND)*

Ellipsoidal region ($\mathcal{G}$). When `r_refine(`ℓ`)` > -1, only cells inside a generalised ellipsoidal region are permitted to refine. The test is

$$\left(\frac{2|\Delta x|}{r}\right)^{e_r} + \left(\frac{2|\Delta y|}{a\,r}\right)^{e_r} + \left(\frac{2|\Delta z|}{b\,r}\right)^{e_r} < 1, \tag{7}$$

where $r =$ `r_refine`, $a =$ `a_refine`, $b =$ `b_refine`, and $e_r =$ `exp_refine`. Cells outside this region are *unflagged* regardless of physical criteria, restricting high resolution to a zoom-in target.

1.1.3. *Cosmological Holdback (NOT)*

Resolution holdback ($\mathcal{H}$). In cosmological runs with cooling, overly eager refinement at high redshift can exhaust memory before the epoch of interest. When `q_refine_holdback=.true.` and the physical cell size $\Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{N_{\rm levelmax}}/a_{\rm exp}$ for levels $\ell >$ `nlevelmax_part`$+3$, the *entire* level is blocked from further refinement:

$$\mathcal{H}(\ell) = \big[\ell > \ell_{\rm part}{+}\ell_{\rm collapse}\big] \wedge \big[\Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{\rm min}/a\big], \tag{8}$$

where $\ell_{\rm part}$ is the particle initial-condition level (typically $\ell_{\rm min}$) and $\ell_{\rm collapse} = 3$. The first holdback level $\ell_{\rm hb}$ is the smallest ℓ satisfying Equation (8):

$$\ell_{\rm hb} = \min\big\{\ell > \ell_{\rm part}{+}\ell_{\rm collapse}\ :\ \Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{\rm min}/a\big\}. \tag{9}$$

1.2. *Buffer Zones*

After applying Equation (1), flagged cells are expanded by `nexpand(`ℓ`)` cell widths in each direction via a dilation operator (`smooth_fine`). This cubic buffer ensures smooth refinement boundaries, preventing the Godunov solver from encountering abrupt level transitions.

2. THE STARBURST PROBLEM

The holdback mechanism blocks *all* refinement criteria at levels $\ell \geq \ell_{\rm hb}$, including the Jeans criterion. This creates two related problems:

Artificial fragmentation.—The Jeans length criterion $\mathcal{J}$ differs fundamentally from the other refinement triggers: it is a *numerical stability* requirement (Truelove et al. 1997), not merely a resolution preference. Violating the Jeans condition leads to artificial fragmentation — a qualitatively wrong result — regardless of epoch. Yet in the standard RAMSES logic (Equation 1), the cosmological holdback $\mathcal{H}$ blocks $\mathcal{J}$ uniformly with all other criteria. At high redshift, dense self-gravitating regions may therefore be held at insufficient resolution, producing spurious fragmentation that propagates into lower-redshift structure.

Abrupt SFR transition.—When the holdback releases level ℓ (i.e. $\mathcal{H}(\ell)$ transitions from true to false), all accumulated criteria fire simultaneously. Many cells that have been gravitationally collapsing during the holdback period now suddenly satisfy the refinement condition, leading to a burst of new cells and an abrupt spike in the star formation rate.

3. JEANS BYPASS: DESIGN AND IMPLEMENTATION

3.1. *Modified Refinement Logic*

The key observation is that the Jeans length criterion is *cell-local*: it depends only on the density and temperature of the cell itself (`uold`), with no neighbor information required. This allows a lightweight evaluation even when the full refinement machinery (which requires neighbor gathering for $\mathcal{D}$, $\mathcal{P}$, $\mathcal{U}$) is skipped by the holdback.

With `jeans_bypass_holdback=.true.`, the refinement logic becomes

$$\mathcal{R}' = \Big[\neg\,\mathcal{H} \wedge \mathcal{G} \wedge \big(\mathcal{M}{\vee}\mathcal{S}{\vee}\mathcal{D}{\vee}\mathcal{P}{\vee}\mathcal{U}{\vee}\mathcal{J}\big)\Big] \vee \Big[\mathcal{H} \wedge \mathcal{J} \wedge (\ell < \ell_{\rm hb}{+}n_{\rm byp} \tag{10}$$

where $n_{\rm bypass} \equiv$ `n_jeans_bypass_levels` (default 1). The geometry mask $\mathcal{G}$ still applies through the non-holdback branch, preserving zoom-in spatial restrictions. Only the temporal holdback $\mathcal{H}$ is bypassed for $\mathcal{J}$, and only within a limited number of levels.

This modification is analogous to the existing treatment of sink particles: $\mathcal{S}$ is already evaluated inside the holdback-protected block, but unlike $\mathcal{J}$ it only triggers in cells that already host a sink. Extending the same exemption to $\mathcal{J}$ ensures numerical integrity for gas that is *about to* form a sink.

3.2. *Level Limitation: Preventing Cascades*

Without a level limit ($n_{\rm bypass} \to \infty$), the Jeans bypass creates a catastrophic cascade. When a cell at the holdback level $\ell_{\rm hb}$ is flagged by the Jeans criterion, child

cells are created at level $\ell_{\rm hb}+1$. Since level $\ell_{\rm hb}+1$ is also in the holdback zone, the bypass applies again, flagging cells at $\ell_{\rm hb}+2$, and so on until $\ell_{\rm max}$. This rapidly exhausts available memory.

The cascade occurs because the Jeans length in radiatively cooling gas (below the star-formation density threshold) shrinks without bound as the temperature drops. There is no physical mechanism to halt the Jeans criterion from triggering at every successive level — only the polytropic temperature floor (Section 4) or a hard level limit can stop it.

Table 1 shows the progression observed in a test run without level limit.

The memory footprint grew from 81% to 88.5% in a single coarse step before crashing. The fine-step timestep also decreased by 8× due to subcycling at levels 15–17.

With $n_{\rm bypass}=1$, only level $\ell_{\rm hb}=14$ is bypassed, creating cells at level 15 but no further. This provides a controlled pre-refinement before the holdback releases.

4. ROLE OF THE POLYTROPIC TEMPERATURE FLOOR

4.1. *Jeans Length in the Polytropic Regime*

In RAMSES, the star-formation module enforces a polytropic equation of state above the density threshold $n_\star$:

$$T(n) = T_{2,\star}\left(\frac{n}{n_\star}\right)^{g_\star-1}, \qquad n \geq n_\star, \tag{11}$$

with default parameters $T_{2,\star}=750\,{\rm K}$, $n_\star=0.1\,{\rm cm}^{-3}$, $g_\star=5/3$. This floor is applied in the cooling routine and is already reflected in `uold`, so the Jeans length computation in `jeans_length_refine()` automatically includes the polytropic stabilisation.

Substituting Equation (11) into Equation (6):

$$\lambda_{\rm J} \propto \sqrt{\frac{T}{\rho}} \propto \sqrt{\rho^{g_\star-1}/\rho} = \rho^{(g_\star-2)/2}. \tag{12}$$

For $g_\star=5/3$: $\lambda_{\rm J}\propto\rho^{-1/6}$. The Jeans length *decreases very slowly* with density — one or two additional levels of refinement are typically sufficient to resolve it.

In contrast, for gas below $n_\star$ (no polytropic floor), radiative cooling can drive T to very low values while ρ remains moderate. This yields $\lambda_{\rm J}\ll\Delta x_\ell$ at many levels, causing the cascade.

4.2. *Self-regulation via `d_jeans_thre`*

The density threshold parameter `d_jeans_thre` (in units of $n_{\rm H}\,[{\rm cm}^{-3}]$) restricts the Jeans criterion to cells above a given density:

$$\mathcal{J}({\rm cell}) = \begin{cases} N_J\,\Delta x_\ell \geq \lambda_{\rm J} & \text{if } n \geq \texttt{d_jeans_thre}, \\ \text{false} & \text{otherwise.} \end{cases} \tag{13}$$

Setting `d_jeans_thre` $\approx n_\star$ restricts the Jeans criterion to the polytropic regime. The cascade then self-limits through the following mechanism:

1. At level $\ell_{\rm hb}$: only cells with $n\geq n_\star$ are checked. The polytropic floor guarantees $T\geq T_{2,\star}$, so $\lambda_{\rm J}$ is relatively large.

2. If $N_J\,\Delta x_{\ell_{\rm hb}} > \lambda_{\rm J}$, the cell is flagged and children at $\ell_{\rm hb}+1$ are created.

3. At level $\ell_{\rm hb}+1$: Δx is halved, density is approximately unchanged, $\lambda_{\rm J}$ is approximately unchanged.

4. $N_J\,\Delta x_{\ell_{\rm hb}+1} = N_J\,\Delta x_{\ell_{\rm hb}}/2 < \lambda_{\rm J} \Rightarrow$ Jeans criterion is *satisfied* $\Rightarrow$ no further flagging $\Rightarrow$ cascade stops naturally.

This provides a *physics-based* self-regulation, complementary to the *hard limit* imposed by `n_jeans_bypass_levels`.

When `d_jeans_thre` $=0$ (default), *all* cells are checked, including low-density gas where cooling produces arbitrarily small $\lambda_{\rm J}$. This is the root cause of the unlimited cascade observed in Section 3.2.

5. IMPLEMENTATION DETAILS

5.1. *Code Structure*

The implementation modifies two source files (plus parameter declaration and namelist reading).

`flag_utils.kjhan.f90` — `userflag_fine()`—The holdback early-return is modified from a simple `return` to a conditional block:

```
if(prevent_refine .and. q_refine_holdback) then
   if(jeans_bypass_holdback .and. hydro &
      .and. n_jeans_bypass_levels > 0) then
      ! Find the first holdback level
      dx_min = (0.5D0**nlevelmax)*scale
      holdback_threshold = &
        (4d0**(1d0/ndim))*(dx_min/aexp)
      holdback_first = nlevelmax + 1
      do jlevel = nlevelmax_part &
         + nlevel_collapse+1, nlevelmax
         dx_test = (0.5d0**jlevel)*scale
         if(dx_test < holdback_threshold) then
            holdback_first = jlevel
            exit
         endif
      end do
      ! Only bypass within allowed range
      if(ilevel < holdback_first &
         + n_jeans_bypass_levels) then
         call jeans_only_flag(ilevel)
      endif
   endif
   return
endif
```

Table 1. Cascade refinement without level limit ($a = 0.164$, $\ell_{\rm hb} = 14$, $\ell_{\rm max} = 17$). All levels ≥ 14 are in the holdback zone.

Level	$\Delta x/\Delta x_{\rm min}$	Holdback	Bypass	$\mathcal{J}$ fires	Result
≤ 12	≥ 32	No	—	Yes (normal)	Normal
13	16	No	—	Yes (normal)	Normal
14	8	Yes	Active	Yes	$\rightarrow$ lv. 15 created
15	4	Yes	Active	Yes	$\rightarrow$ lv. 16 created
16	2	Yes	Active	Yes	$\rightarrow$ lv. 17 created
17	1	$\ell_{\rm max}$	—	—	OOM crash

Table 2. Two complementary mechanisms for preventing the bypass cascade.

Mechanism	Type	Behaviour
`n_jeans_bypass_levels`	Hard limit	Caps extra levels regardless of physics
`d_jeans_thre` $\approx n_\star$	Physics-based	Polytropic floor limits $\lambda_{\rm J}$ decrease

hydro_flag.kjhan.f90 — *jeans_only_flag()*—A new lightweight subroutine that:

1. Loops over active grids with OpenMP parallelism.
2. Computes cell indices without neighbor gathering.
3. Calls the existing `jeans_length_refine()` subroutine (no code duplication).
4. Sets `flag1` for flagged cells and updates `nflag`.

This is possible because `jeans_length_refine()` only accesses cell-local data (`uold(ind_cell,...)`), unlike the gradient-based criteria $\mathcal{D}$, $\mathcal{P}$, $\mathcal{U}$.

5.2. *Safety Properties*

- **Default off**: With `jeans_bypass_holdback=.false.` (default), the new code path is never entered. Bit-identical to the unmodified code.
- **No neighbor access**: `jeans_only_flag()` is safe to call at any point in the refinement cycle.
- **OMP safety**: `flag1` writes are grid-partitioned; `nflag` uses reduction.
- **Hard level limit**: `n_jeans_bypass_levels` provides a fail-safe even if `d_jeans_thre` is misconfigured.

6. SUMMARY OF PARAMETERS

Table 3 lists all refinement-related namelist parameters, including the new bypass parameters.

The recommended configuration for cosmological runs with star formation is:

```
&REFINE_PARAMS
m_refine           = 9*8.,
jeans_refine       = 9*4.,
jeans_bypass_holdback  = .true.
n_jeans_bypass_levels  = 1
d_jeans_thre       = 0.1      ! = n_star
q_refine_holdback = .true.
/
```

7. TEST RESULTS

All tests use the HR5 medium-box setup: $\ell_{\rm min} = 9$, $\ell_{\rm max} = 17$, 12 MPI ranks, OMP=5, restarting from output 5 ($a = 0.164$, $z \approx 5.1$, nstep_coarse=241).

7.1. *Bit-identical Verification*

With `jeans_bypass_holdback=.false.` (default) and no `jeans_refine`, the new binary produces results identical to the reference run: econs=6.13×10^{-4} at step 242, econs=6.71×10^{-4} at step 243.

7.2. *Unlimited Bypass*

With `jeans_refine=9*4.`, `jeans_bypass_holdback=.true.`, `d_jeans_thre=0`, and no level limit: cascade refinement from level 14 through 17 within a single coarse step. Memory grew from 81% to 88.5% before OOM crash. Fine-step timestep decreased by 8× due to subcycling.

Table 3. AMR refinement parameters in the `&AMR_PARAMS` and `&REFINE_PARAMS` namelists. New parameters are marked with ⋆.

Parameter	Default	Role
`m_refine`	8.0	Lagrangian mass threshold (per level)
`sink_refine`	F	Force max level around sinks
`err_grad_d`	−1	Density gradient threshold
`err_grad_p`	−1	Pressure gradient threshold
`err_grad_u`	−1	Velocity gradient threshold
`jeans_refine`	−1	Jeans length in cells (per level)
`d_jeans_thre`	0	Density threshold for Jeans ($n_{\rm H}\,{\rm cm}^{-3}$)
`r_refine`	−1	Geometry region diameter (per level)
`a_refine`	1.0	Ellipticity Y/X
`b_refine`	1.0	Ellipticity Z/X
`exp_refine`	2.0	Norm exponent
`x/y/z_refine`	0.5	Region centre
`nexpand`	1	Buffer zone width (per level)
`q_refine_holdback`	F	Cosmological holdback
⋆ `jeans_bypass_holdback`	F	Enable Jeans bypass of holdback
⋆ `n_jeans_bypass_levels`	1	Max extra levels beyond $\ell_{\rm hb}$

7.3. *Limited Bypass ($n_{\rm bypass} = 1$)*

With `n_jeans_bypass_levels=1`: only level 14 is bypassed, cells at level 15 may be created but no further cascade.

7.4. *Future: `d_jeans_thre` $= n_\star$*

A test with `d_jeans_thre=0.1` will verify the physics-based self-regulation, potentially allowing $n_{\rm bypass} > 1$ without cascade, since only gas in the polytropic regime would trigger the Jeans criterion.

8. DISCUSSION

8.1. *Physical Motivation*

The holdback mechanism was designed to prevent over-refinement from the mass criterion at early times. However, it inadvertently blocks the Jeans criterion, which has a different physical motivation: resolving the Jeans length prevents artificial fragmentation and ensures numerically converged star formation.

By allowing the Jeans criterion to operate ahead of the holdback release, we expect:

- Dense gas that will form stars is pre-refined before the holdback releases.
- When the holdback releases, the transition is smoother because the densest regions are already at the appropriate resolution.
- The SFR curve shows a gradual onset rather than an abrupt step.

8.2. *Interplay of the Three Parameters*

The bypass is controlled by three parameters that form a layered defence:

Layer 1: `jeans_bypass_holdback` (on/off)—Master switch. When off (default), the system is identical to standard RAMSES.

Layer 2: `d_jeans_thre` (physics-based regulation)—Restricts the Jeans check to dense gas where the polytropic floor operates. In this regime, $\lambda_{\rm J} \propto \rho^{-1/6}$ varies slowly, and 1–2 extra refinement levels suffice to resolve it. Provides self-consistent, physics-driven cascade termination.

Layer 3: `n_jeans_bypass_levels` (hard safety limit)—Absolute cap on the number of extra levels. Prevents OOM regardless of physics configuration. Essential when `d_jeans_thre=0` (all cells checked).

REFERENCES

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